

Probability

predictions the result
of unknown events

chance

uncertainty
(Bilinmezlik)

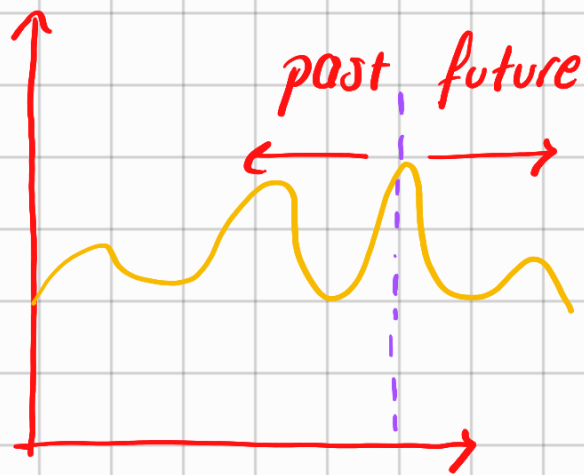
olasılıkları tanımla-
mak için kulları-
dığımız terimler
bunlar

→ mesela ben yazı
tura otıyorum
ama ben bu olayı
yaparken bütün
parametreleri bilsem
mesela parmanın kırıveti
hızının direnci böyle
olsaydı ben bu olayı
hesaplayabilirdim

Weather conditions - rain at 14.15 pm in
Darıutpasa?

→ forecasts 80% rain 14-15 pm

Stock prices



Example: flipping a coin

1- Head

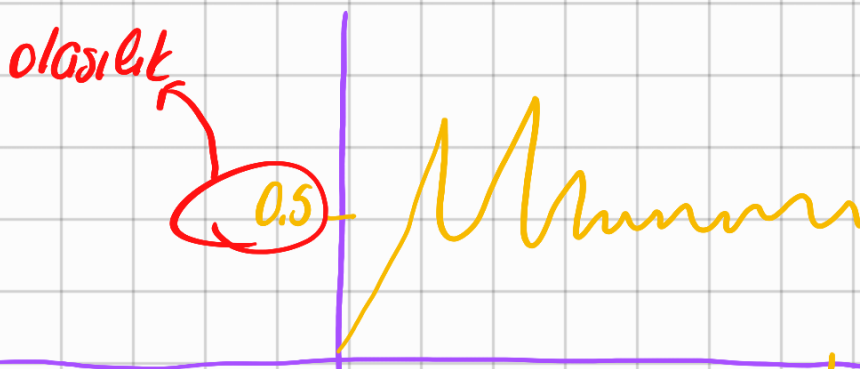
2- Tail

If I toss a coin what is the probability of heads?

$\{H, T\}$

$$P(T) = 1/2 = 50\% = 0.5$$

$$P(H) = 1/2 = 50\% = 0.5$$



100 → Kaç
kere
yazı tura
atıldığı

if we have N equally likely outcomes

$$P(\text{each}) = 1/N$$

Sets

A set is an unordered collection of things (elements)

$$A = \{1, 2, 3, 3\} \quad B = \{1, 2, 3\}$$

Burdaki A ve B kümesi birbirine eşittir

Kümelerin içinde sayılar, semboller, karakterler olabilir

Subset (altküme)

set A is a subset of B if every element of A is also an element of B

$$A \subset B \equiv (x \in A) \Rightarrow (x \in B)$$

↪ x bir a küme-

sinin elemanında
o zaman b kümesinin
bir elemanı olmalıdır

C (subset)

$$A = \{x^3; x=2,3,4\} = \{2^3, 3^3, 4^3\}$$

→ curly braces

→ such that

→ List

$$N_{\text{natural numbers}} = \{1, 2, 3, \dots\}$$

$$Z = \{\dots -1, 0, 1, 2 \dots\}$$

tam sayılar (integers)

→ oran

Rational (ratio)

→ kesirli sayılar

a/b

Symbol is $\mathbb{Q}$

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a \in \mathbb{Z} \text{ and } b \in \mathbb{Z} - \{0\} \right\}$$

Real
numbers

$\hookrightarrow$ infinite
numbers

$\mathbb{R} \rightarrow$ Real
numbers

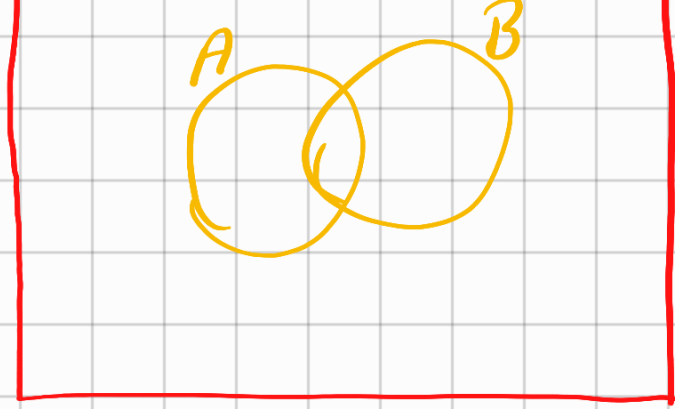
SuperSet ' $\supset$ '

If A is a subset of set B , then B is
superSet of A

Kümeleri listelerle gösterebiliriz
bir de venn diagramları ile
gösterebiliriz

Venn diagrams

Universal



Set
 $\rightarrow \mathcal{S}, \Omega$
 (omega)

$A \subset \Omega$ true

Set operations

Union (Birleşim) $\rightarrow$ the union of two sets
 A and B consists of
 all elements in A or B

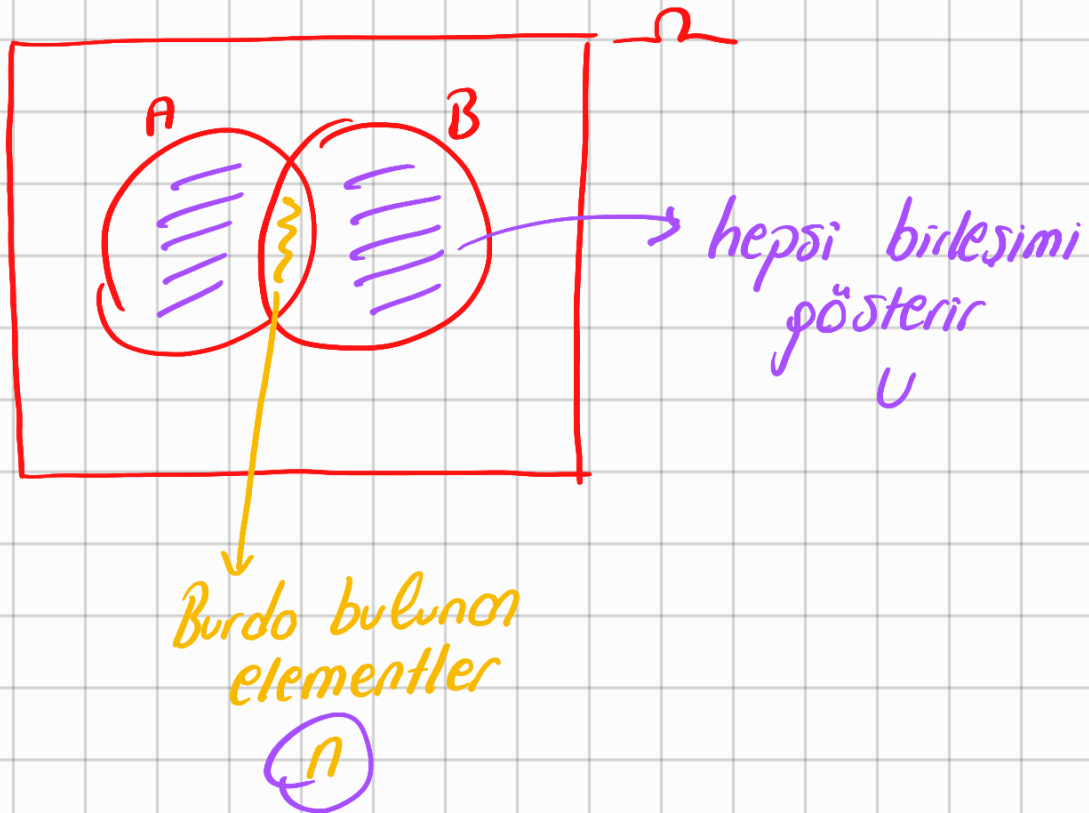
$$x \in (A \cup B) \iff (x \in A) \text{ or } (x \in B)$$

$\downarrow$ $\downarrow$ $\downarrow$
 $\{1, 2\}$ $\{4\}$ Ancak ve ancak
 $A \cup B = \{1, 2, 4\}$
 Union'un matematiksel tanımı

Intersection (Kesişim)

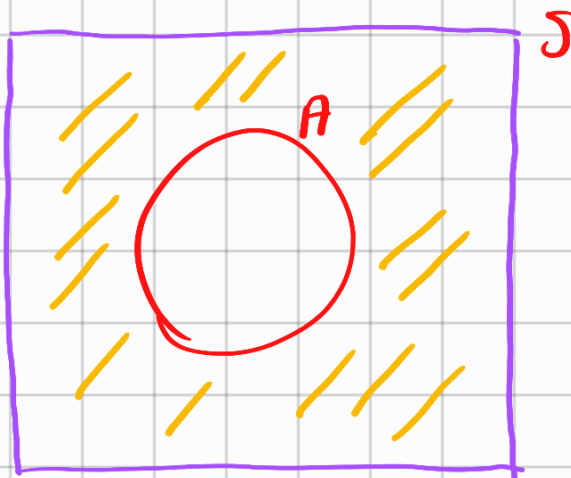
The intersection of two
 sets A and B consists
 of all elements in both

of two elements in both
A and B



Complement

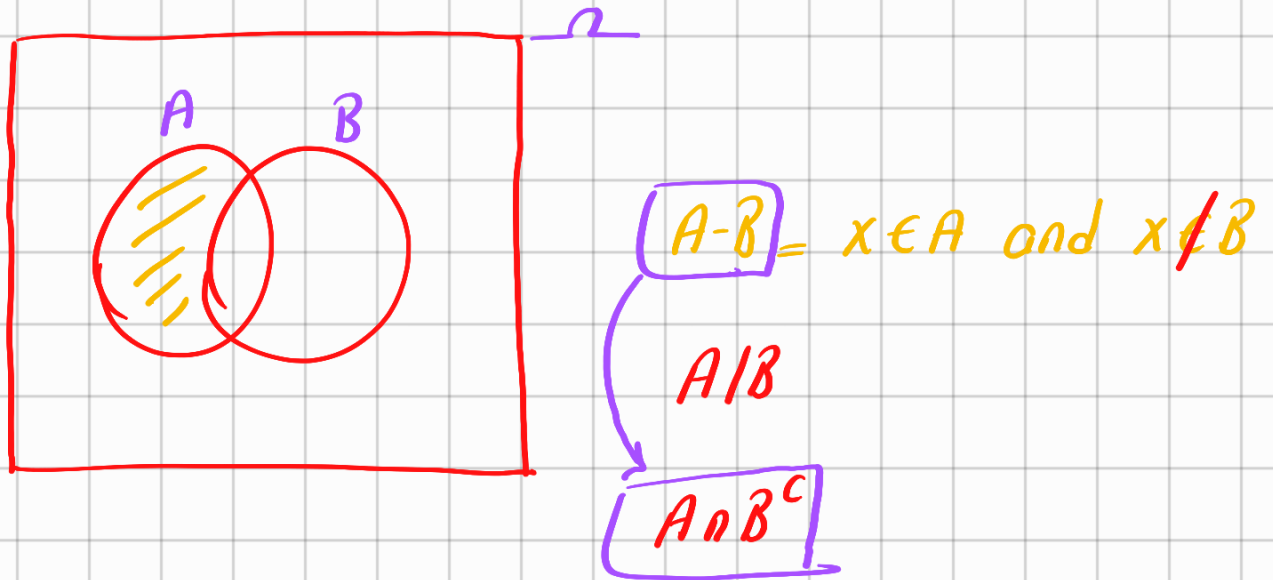
the complement of a set A is set of all elements in Ω that are not in A



$A^c, A', \bar{A}$

Difference (subtraction)

The difference of a set B from A is all elements in A that are not in B



Disjoint sets (Ayrık küme) (Mutually exclusive set)

Two sets A and B are disjoint $A \cap B = \emptyset$

$\emptyset \rightarrow$ empty set (Boş küme)
elemanı olmayan bir kümedir





are disjoint set

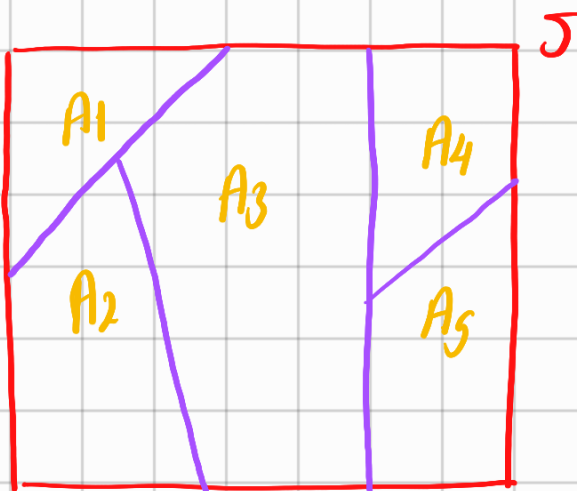
Partition:

A collection of sets;

$A_1, A_2, \dots, A_n$ is a partition set if and only if

a- They are disjoint

b- $A_1 \cup A_2 \cup A_3 \dots \cup A_n = S$



→ Biz o zaman
şöyle diye-
biliriz $A_1, A_2, \dots, A_5$
partition set'tir

De Morgan's law (De Morgan kanunları)

$$(A \cup B)^c = A^c \cap B^c$$

Example

$$S = \{1, 2, 3, 4, 5, 6\} \quad A = \{1, 2\} \quad B = \{2, 4, 6\}$$

$$A \cup B = \{1, 2, 4, 6\}$$

$$A \cap B = \{2\}$$

$$A^c = \{3, 4, 5, 6\}$$

$$B^c = \{1, 3, 5\}$$

$$(A \cup B)^c = \{1, 2, 4, 6\}^c = \{3, 5\}$$

De Morgan's law

$$A^c \cap B^c$$

$$\{3, 4, 5, 6\} \cap \{1, 3, 5\}$$

$$\hookrightarrow \{3, 5\} \quad \text{proof?}$$

Therom : Distribute law

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$4 * (2 + 17) \rightarrow (4 * 2) + (4 * 17)$$

Functions

Fonksiyonlar bir kümeden diğerine eşlemedir
bir fonksiyonları nasıl gösteririz

fonksiyonlara genellikle f ismini veririz

$$f: X \rightarrow Y$$

domain $\rightarrow$ co-domain

$$\forall x \in X, f(x) \in Y$$

Mesela bu tanımlı
değil 0 zaman
napıyoruz $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$

Böyle
yapıyoruz



domain

co-domain

Range: the set of all possible values of $f(x)$
(x 'in tüm olası değerlerin kümesidir)

$$\text{Range} \subset Y$$

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ defined as } f(x) = x^2$$

diyelim ki gerçel sayılardan gerçel sayılara f denilen fonksiyonumuz var

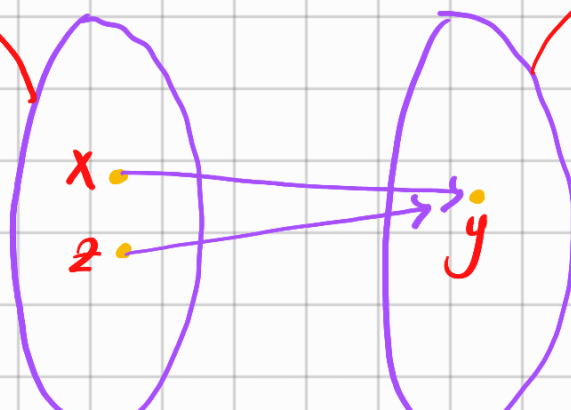
$$X = Y = \mathbb{R}$$

$$\text{range}(f) = \mathbb{R}^+ = \{x \in \mathbb{R} \mid x \geq 0\}$$

Joru $\rightarrow$ x 'teki 2 element
 y 'deki tek elemente
gidebilir mi

Gocuk

Anne



$\rightarrow$ 2 Gocugun bir annesi olabilir
ama bir Gocugun 2 annesi olamaz

$f(x)=y$
 $f(2)=y$ } Böyle bir şey
olabilir mi

Olabilir

Yoru → x 'teki bir
element y 'deki 2 element ile
eşleşebilir mi

$f(2)=y$
 $f(2)=w$ } Bu olabilir mi?

Sadece bir
Yoru olmak
bu belirsizlik
anlamına
gelir

Olmaz

Countable and Uncountable
sets

finite set

Yarlıkların tüm unsurları biliniyorsa

$$A = \{1, 2, 4, 8, 9\}$$

$$x \in [4, 8) \text{ } \delta \text{ } x \in 2$$



$$4, 5, 6, 7$$

infinite set

$$\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$$

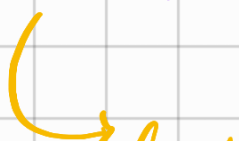
Countable and Uncountable sets

$|A| \rightarrow$ cardinality

$$A = \{1, 2, 3\}$$

$$|A| \rightarrow 3$$

A set is finite if $|A| < \infty$



finite set

A set Countable if it is finite or its elements can be enumerated or listed in a sequence

$$A = \bigcup_{k=1}^{\infty} \{a_k\}$$

→ its means

$$A = \{a_1, a_2, a_3, \dots\}$$

Let's Natural numbers

$$\hookrightarrow \{1, 2, 3, \dots\}$$

it is an infinite set

It countable ?

→ Bir kümenin elemanları sayılabilir ise countable olur

→ Yes it is countable

Uncountable sets

the element can not be enumerated

$\mathbb{R} \rightarrow$ real number

$\rightarrow$ Real numbers is uncountable set
(Real sayıları sayamadığın için uncountable olur)

$\rightarrow$ No enumeration

A set is countable infinite if it is in one to one correspondence with

$$\mathbb{N} = \{1, 2, 3, \dots\} = \bigcup_{k=1}^{\infty} \{k\}$$

$\mathbb{Z} \rightarrow$ countable ?

$$\mathbb{Z} \rightarrow \{ \dots, 0, 1, 2, \dots \}$$

$a_0 \quad a_1 \quad a_2$

$\mathbb{Z} \rightarrow$ infinite sets but it is countable sets

We can enumerate them

$\mathbb{Q}^+$ (Rational numbers)

$$\rightarrow \left\{ \frac{a}{b} : a \text{ or } b > 0 \right\}$$

$\swarrow$
infinite set

$$\mathbb{Q}^+ = \bigcup_{i,j=1}^{\infty} \left\{ \frac{i}{j} \right\} \rightarrow \bigcup_{i=1}^{\infty} \bigcup_{j=1}^{\infty}$$

$\frac{i}{j} \rightarrow$ Burası 2'ye
geçtiği zaman
aşağısı 1'den
tek-
rar
başlıyor

	0	1	2	3	4	5
0						
1	$0/1$					
2	$0/2$	$1/2$				
3		$1/3$				
4						
5						

I can enumerate them

(reel sayilar)

$\mathbb{R} \rightarrow$ infinite $\rightarrow$ Yes

Countable No

$\mathbb{R}$ is not countable

$$[a, b] \quad b > a$$

We can not enumerate them
so it is uncountable

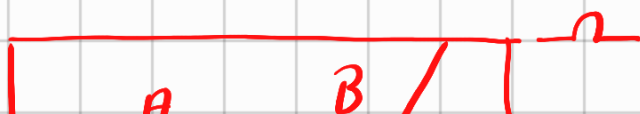
Power Set

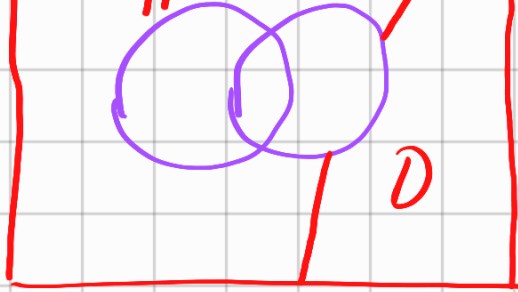
Set of all subsets of a set A

$$A = \{1, 2\}$$

$$P(A) = \{\{1\}, \{2\}, \{1, 2\}, \emptyset\}$$

Exhaustive Sets





Sets A, B, C, D
are exhaustive
if their union is
 Ω

$$A \cup B \cup C \cup D = \Omega$$

intersections are allowed

Probability theory

① random experiment $\rightarrow$ Bir sonucu olmayan fenomendir

A phenomenon whose
outcome cannot be
predicted with certainty

- Rolling a die
 - Flipping a coin
 - Rolling a die 3 times
- } random experiment

② Outcome :

outcome is a result of random exp.

→ Rolling a die → 1, 2, 3, 4, 5, 6

→ 3



flipping a coin $\rightarrow H$

Roll o die 2 times $(6, 2)$
(20r atmak)

③ **Event**
(olası sonuçların bir toplamıdır)

An event is a collection of possible outcomes

Rolling a die

$$E_1 = \{1, 3, 5\}$$

$$E_2 = \{2, 4\}$$

④ Sample space

Set of possible outcomes

↓
Ω (Bununla gösterilebilir)

Rolling a die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$\Omega = \{H, T\}$$

Roll a die 3 times

$$\Omega = \{(1,1,1), (1,1,2), (1,1,3), \dots, (6,6,6)\}$$

↓
all of possible outcomes

↙
6 x 36 possible outcomes

6 6 6

↘
 $\frac{216}{7}$

↘
3. yer için
6 seçenek
var

Birinci
yer için
6 seçenek
var

2. yer için
6 seçenek
var

Random experiment sonucu olmayan bir fenomendir

Sample set tüm sonuçları içeren bir kümedir

→ Event is a set

Sample space = universal set

Outcome → element of a set

Partition

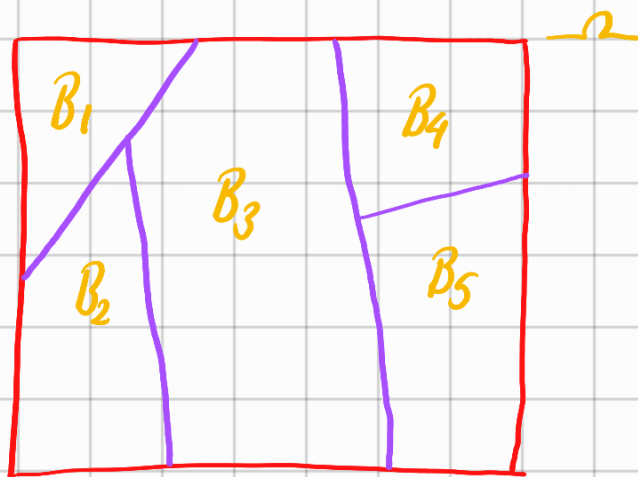
A partition is collectively exclusive and mutually exclusive set of events

$A_1, A_2, \dots, A_n$ is a partition

$A_1 \cup A_2 \cup \dots \cup A_n = \Omega$

$$A_1 \cup A_2 \cup \dots \cup A_n = \Omega$$

$$A_i \cap A_j = \emptyset, i \neq j$$



Sample space $\rightarrow$ Tüm olası sonuçlardır

$\hookrightarrow$ universal
sete çok
benzer

event sample space ve universal setin alt kümesidir

We say that an event A is occurred if the outcome of the experiment is an element of A

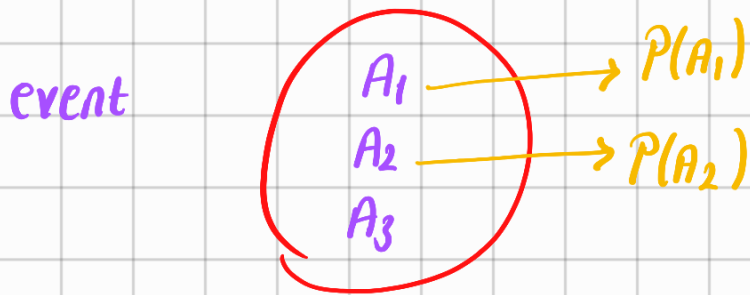
Probability

An event $A \rightarrow P(A)$ Probability of A

$\nearrow$ Buna o olasılığı

denir

We assign a probability $P(A)$ to every event A



Axioms of probability

$P(\cdot)$ is a function that maps events in the sample space Ω to real numbers

- ① for any event A $P(A) \geq 0$
- ② Probability of the sample space Ω $P(\Omega) = 1$
- ③ for any countable collection $A_1, A_2, \dots$ of disjoint events

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

